CAPSTONE PROJECT

Canvas Helper AI

Intelligent Assistant for Canvas LMS Administration

Team 16 · COSC 499 Capstone
Client: Mohamed Mostafa

Our Team



Jessica Goel

Frontend Developer



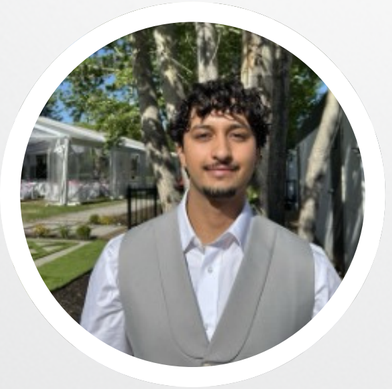
Mridul Nohria

Integration Lead / DevOps / QA Lead



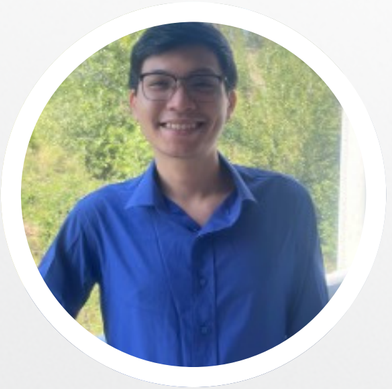
Yerdana Maulenbay

Backend Developer



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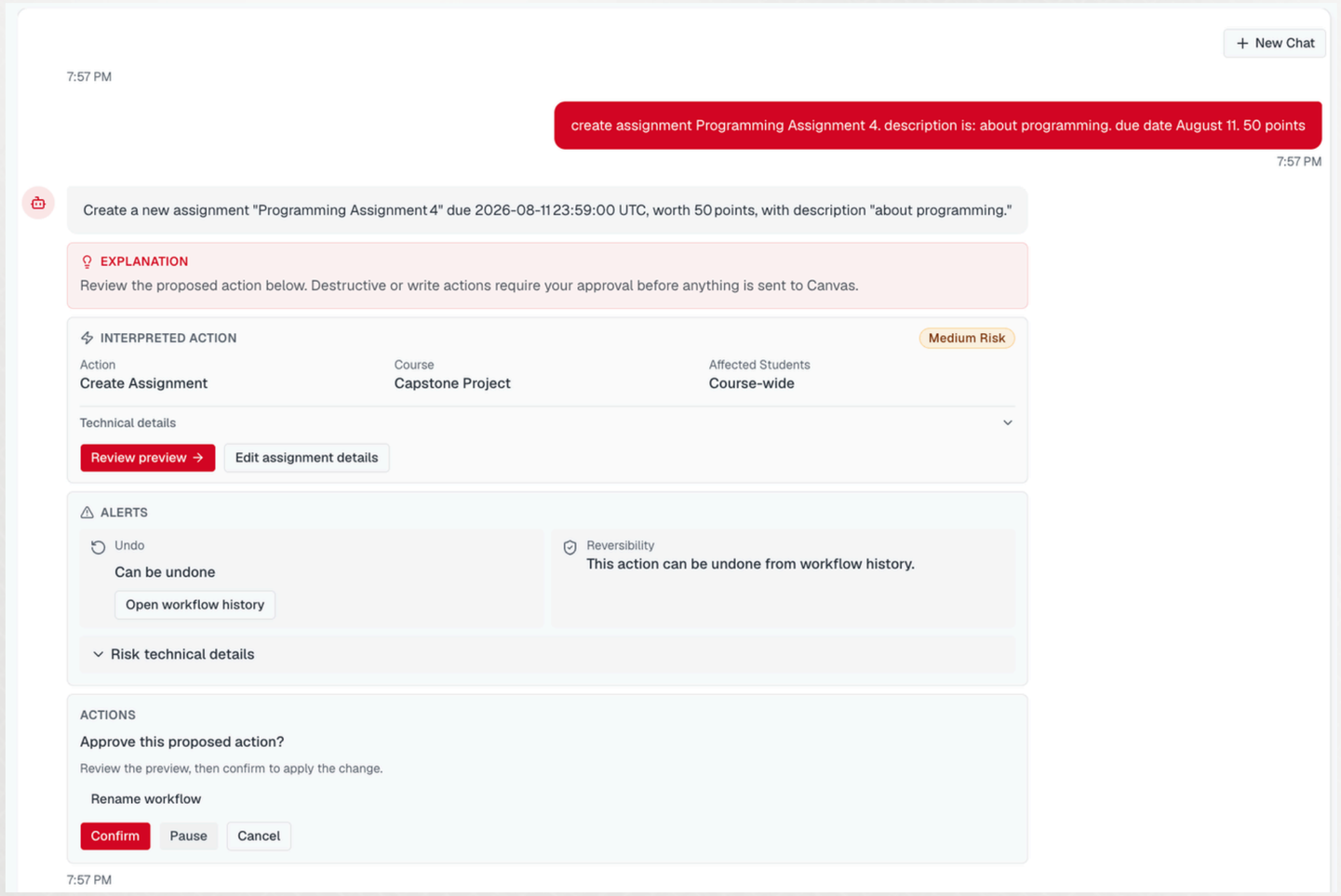
THE PROBLEM



"one routine update = many places"

Canvas Admin Work is Slow and Fragmented

- Too many screens for routine tasks
- Autonomous AI writing to Canvas is unsafe
- Fragmented workflows slow down instructors
- No confirmation layer for high-risk actions



UVP & USERS

Value Proposition

- Natural-language Canvas ops
- Preview → approve every write
- High-risk needs typed CONFIRM
- Encrypted tokens; never sent to LLM

Target Users:

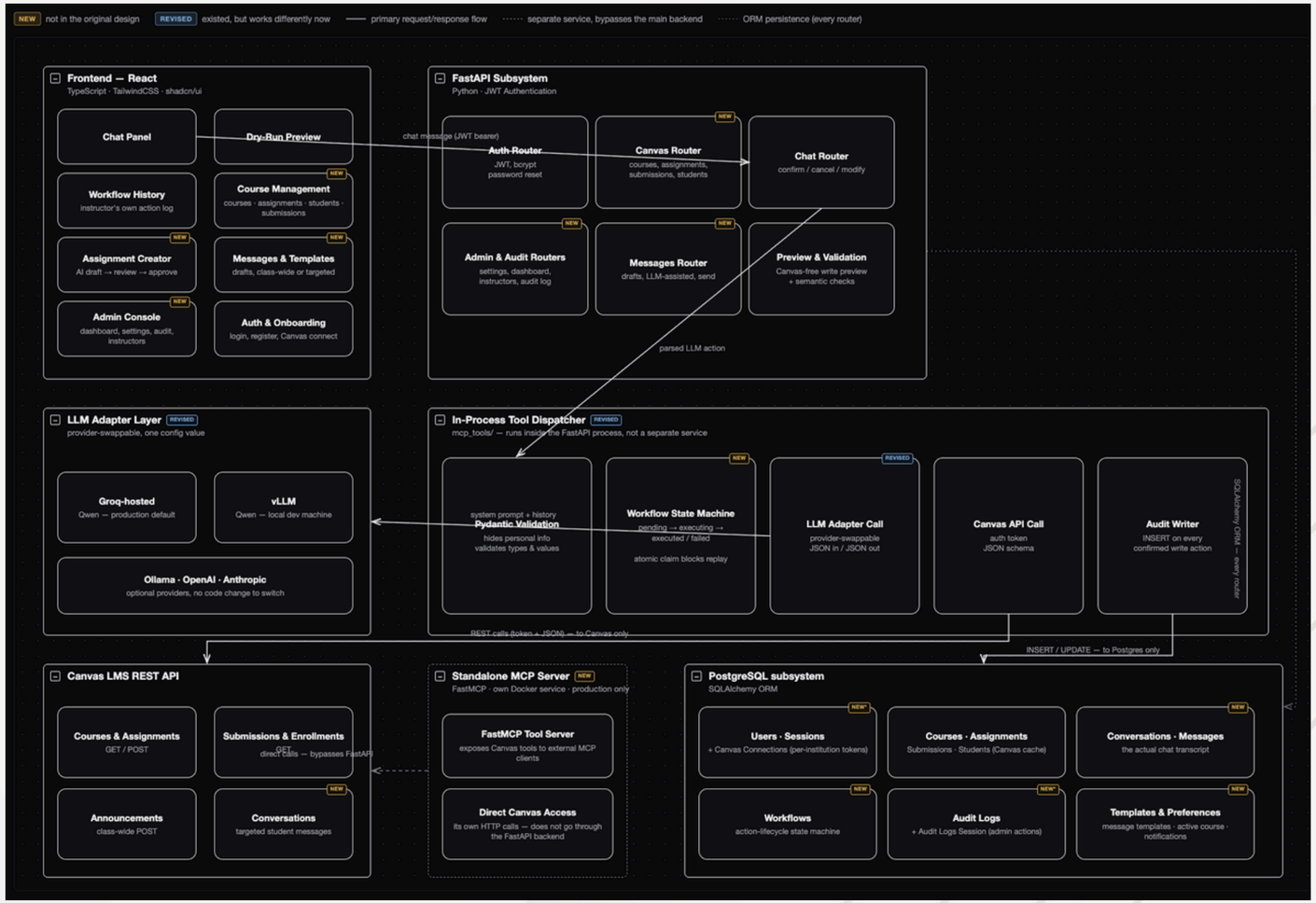
Instructors (primary)

Admins (secondary)

ARCHITECTURE OVERVIEW

Component Flow

React → FastAPI → Services / Validation → Canvas API
FastAPI ↔ PostgreSQL
FastAPI ↔ LLM Providers
Optional MCP boundary



PLAN → FINAL

Planned vs. Shipped

A comparison of what was originally planned at the start of the project versus what was ultimately shipped in the final product.

01

Local LLM-first → Multi-provider AI

02

MCP-centric → Optional MCP boundary

03

Monolithic → Modular services

04

Preview only → Risk tiers + CONFIRM

Two Rounds of User Testing

R1: AI rough; preview/confirm loved; errors unclear.
R2 (34 users): ~62% rated usability 4–5; nav strong; feedback shifted to polish.

UX TESTING

- **~62% usability score (4–5 / 5)**
- **Nav strong · feedback shifted to polish**
- **Faster AI · context sync · clearer errors**
- **Loading & chat UX improvements shipped**

TESTING RESULTS

01 Pytest 1,297 · 78.6%

02 Vitest 1,075 · 86.1%

03 Playwright 16/16 · Fuzz 9/9

04 36/36 requirements PASS

Layered unit → E2E → UX → acceptance

All test suites passed across unit, integration, end-to-end, and acceptance layers. Full coverage validates safety-critical confirm gates, Canvas operations, and UI flows.

CHALLENGE + LESSONS

Key Challenges & What We Learned

The Core Challenge

Helpful AI without unsafe or duplicate Canvas writes — balancing capability with strict guardrails was the hardest design constraint throughout the project.

Our Solution Approach

Server-side grounding, confirm gates before every write, and idempotency checks to prevent duplicate Canvas operations.

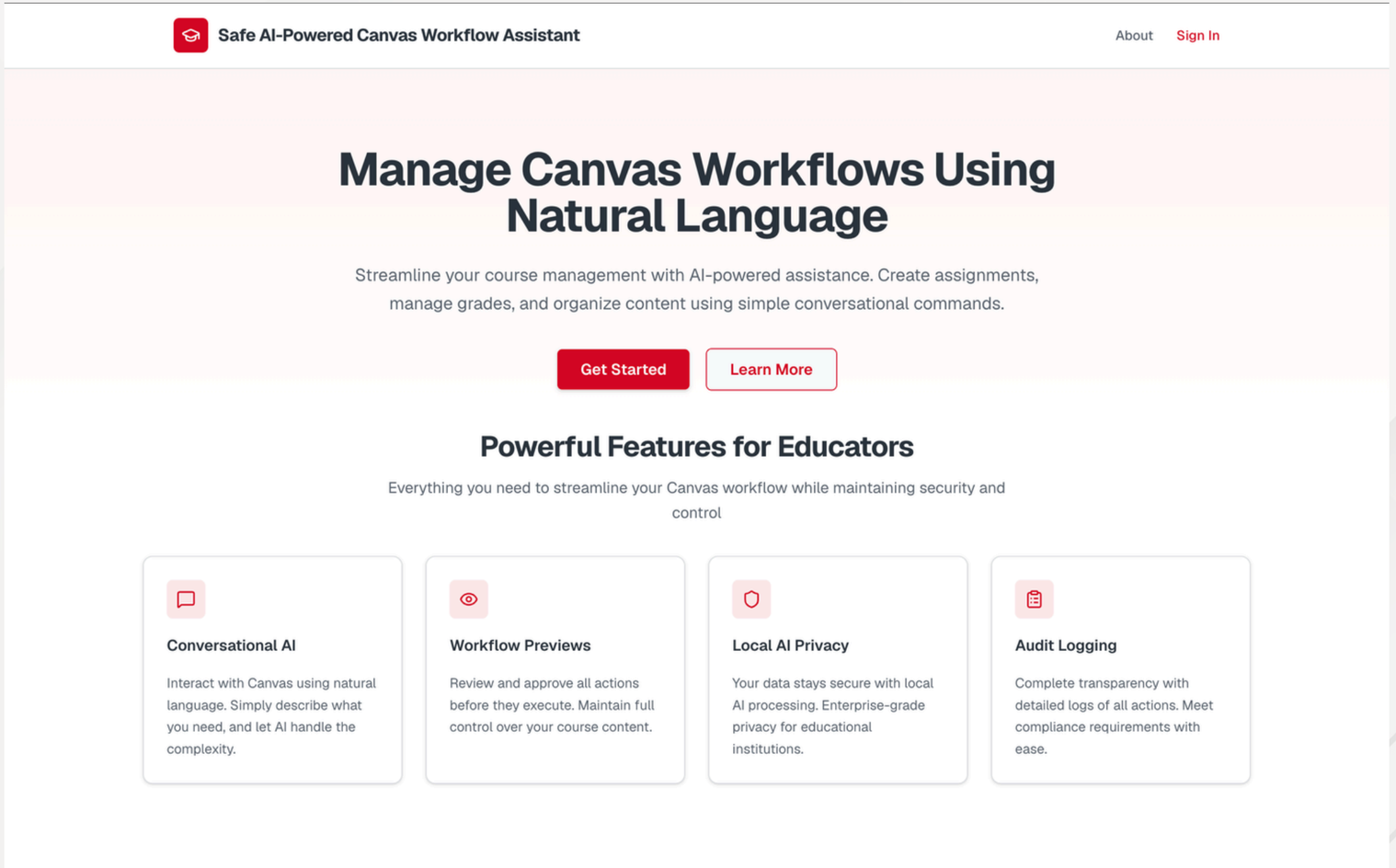
Lesson 1: Safety & UX First

Keep safety-first principles and run UX testing rounds early and continuously — user trust depends on predictable, transparent AI behavior.

Lesson 2: Model & Context Strategy

Pin models and implement context sync earlier in the development cycle to avoid costly refactors and inconsistent AI responses.

LIVE DEMO


Safe AI-Powered Canvas Workflow Assistant
About
Sign In
Manage Canvas Workflows Using Natural Language
Streamline your course management with AI-powered assistance. Create assignments, manage grades, and organize content using simple conversational commands.
Get Started
Learn More
Powerful Features for Educators
Everything you need to streamline your Canvas workflow while maintaining security and control
Conversational AI
Interact with Canvas using natural language. Simply describe what you need, and let AI handle the complexity.
Workflow Previews
Review and approve all actions before they execute. Maintain full control over your course content.
Local AI Privacy
Your data stays secure with local AI processing. Enterprise-grade privacy for educational institutions.
Audit Logging
Complete transparency with detailed logs of all actions. Meet compliance requirements with ease.

QUESTIONS?

Open Floor for Discussion

We welcome your questions, feedback, and ideas.

Team 16 · COSC 499

THANK YOU

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